CS 520: Project Requirements

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Team 25
Doc link: https://docs.google.com/document/d/1n_6oGmbsDp2-PY43ljyQBS3wxNS6LW3M/edit

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1. Requirements

1.1 Overview

Finding safe, affordable, and convenient off-campus housing is one of the most stressful challenges faced by UMass Amherst students. Current solutions are fragmented across general-purpose platforms such as Craigslist or Facebook Marketplace, which lack student-specific trust signals and community features. CampusNest is a dedicated web application designed to centralize the off-campus housing experience for UMass Amherst and Five College students. The platform enables students to browse verified housing listings near campus, evaluate poster credibility through linked social media profiles (Facebook, Instagram, and Snapchat), and connect with compatible roommates — all within a single student-focused environment.

The primary stakeholders are UMass Amherst undergraduate and graduate students who are seeking off-campus housing or roommates, as well as students who wish to sublet their apartments. The system aims to reduce housing search friction, improve listing transparency through social verification, and foster a trusted peer community. By serving the Five College community, CampusNest addresses a real and recurring need for students across UMass Amherst, Amherst College, Hampshire College, Mount Holyoke College, and Smith College.

1.2 Features

- Student Authentication & Profile Management — Students register using their university email. Profiles include personal details, preferences, and linked social media accounts for trust verification.

- Housing Listings Board — Students can post, browse, search, and filter off-campus housing listings by price, location, bedroom count, and availability. Each listing includes photos, descriptions, and poster profile badges.
- Social Media Integration & Trust Badges — Users can connect their Facebook, Instagram, and Snapchat accounts to their profile. Verified social connections are displayed as trust badges on listings and roommate profiles, increasing credibility.
- Roommate Finder — Students can create a roommate profile listing their lifestyle preferences, budget, and availability. Other students can browse and filter these profiles and initiate contact through in-app messaging.
- Saved Listings & Dashboard — Users can save listings and roommate profiles to a personal dashboard for later review, with the ability to add notes and track housing options.
- Reporting & Moderation — Users can report suspicious or inappropriate listings and profiles. An admin panel allows moderators to review flagged content and take appropriate action.

1.3 Functional Requirements (Use Cases)

The following use cases describe the primary interactions users will have with CampusNest. Each use case is traceable to one or more system features described in Section 1.2.

Use Case ID	UC-01
Use Case Name	Student Registration
Actor(s)	Prospective student user
Trigger	User navigates to the app and clicks 'Sign Up'
Main Success Scenario	User enters their UMass email, creates a password, and verifies their student status. The system sends a verification email. Upon confirmation, the account is activated and the user is redirected to their profile setup.
Failure Cases	Invalid or non-UMass email entered; verification email not received; email already registered.

Use Case ID	UC-02
Use Case Name	Social Media Account Linking
Actor(s)	Registered student
Trigger	User accesses 'Connect Social Accounts' from their profile settings
Main Success Scenario	User selects Facebook, Instagram, or Snapchat. The system initiates OAuth flow. Upon authorization, the linked account's public profile info (name, profile photo) is associated with the user's housing profile to increase trust.
Failure Cases	OAuth authorization denied by user; social platform API unavailable; account already linked to another user.

Use Case ID	UC-03
Use Case Name	Create Housing Listing
Actor(s)	Student landlord / sublet poster
Trigger	User clicks 'Post a Listing' from the homepage
Main Success Scenario	User fills out a form with address, rent, available dates, number of bedrooms/bathrooms, photos, and description. System validates and saves the listing, which is then visible to all registered users.
Failure Cases	Missing required fields (address, rent, dates); uploaded photos exceed size limit; user's account not verified.

Use Case ID	UC-04
Use Case Name	Search & Filter Housing Listings
Actor(s)	Student looking for housing
Trigger	User enters search criteria on the Browse Listings page
Main Success Scenario	User filters listings by price range, distance from UMass campus, number of bedrooms, and availability date. The system returns a sorted list of matching listings displayed on a map and as cards.
Failure Cases	No listings match the given filters; map service unavailable.

Use Case ID	UC-05
Use Case Name	View Listing Details
Actor(s)	Student looking for housing
Trigger	User clicks on a listing card from the search results
Main Success Scenario	System displays full listing details including photos, amenities, rent breakdown, landlord/poster profile with linked social media badges, and a contact button.
Failure Cases	Listing has been taken down since search; images fail to load.

Use Case ID	UC-06
Use Case Name	Create Roommate Profile
Actor(s)	Student looking for a roommate
Trigger	User navigates to 'Find a Roommate' and clicks 'Create Roommate Profile'
Main Success Scenario	User fills out preferences including budget, move-in date, lifestyle habits (sleep schedule, cleanliness, noise tolerance), major, and a short bio. Profile is saved and becomes visible to other students searching for roommates.
Failure Cases	Required fields left blank; uploaded profile picture invalid format.

Use Case ID	UC-07
Use Case Name	Browse & Contact Potential Roommates
Actor(s)	Student looking for a roommate
Trigger	User navigates to the Roommate Finder page
Main Success Scenario	User browses roommate profiles filtered by budget range, move-in date, and lifestyle preferences. User can view a profile's social media badges and send an in-app message or contact request to a potential roommate.
Failure Cases	No matching roommate profiles found; messaging service unavailable.

Use Case ID	UC-08
Use Case Name	Save & Manage Favorite Listings
Actor(s)	Student looking for housing
Trigger	User clicks the 'Save' icon on a listing or roommate profile
Main Success Scenario	The listing or roommate profile is added to the user's saved items list. User can later access all saved items from their dashboard, remove items, or add notes.
Failure Cases	User not logged in when attempting to save; saved item limit reached.

Use Case ID	UC-09
Use Case Name	Report Inappropriate Content
Actor(s)	Any registered student
Trigger	User clicks 'Report' on a listing or profile
Main Success Scenario	User selects a reason (e.g., fraudulent listing, inappropriate content, fake profile) and submits the report. System logs the report and sends a notification to admins. The reported item is flagged for review.
Failure Cases	Report submission fails due to server error; user attempts to report own content.

Use Case ID	UC-10
Use Case Name	Admin Review & Remove Flagged Content
Actor(s)	System administrator
Trigger	Admin logs into the admin dashboard and sees pending flagged items
Main Success Scenario	Admin reviews flagged listings or profiles, views supporting evidence, and chooses to remove the content or dismiss the report. The original poster is notified of any removal action.

Failure Cases	Admin account access issues; flagged item already resolved by another admin.

1.4 Non-Functional Requirements

NFR-01: Performance

The system shall support a minimum of 500 concurrent users without degradation in response time. All page load times shall not exceed 3 seconds under normal load conditions, verified through load testing tools such as Apache JMeter or Locust.

NFR-02: Security & Data Privacy

All user data, including passwords and social media tokens, shall be encrypted at rest using AES-256 and in transit using HTTPS/TLS 1.2 or higher. OAuth 2.0 must be used for all third-party social media authentication flows. User passwords must never be stored in plaintext.

NFR-03: Usability & Accessibility

The application shall meet WCAG 2.1 Level AA accessibility standards, including keyboard navigability and screen reader support. A usability study with at least 5 student participants shall confirm that a first-time user can successfully find and save a housing listing within 5 minutes of onboarding.

NFR-04: Availability & Reliability

The system shall maintain a minimum uptime of 99% during the academic semester (September through May), excluding scheduled maintenance windows. All critical failures shall trigger automated alerts to the development team within 5 minutes.

1.5 Challenges & Risks

Risk 1: Social Media API Instability

Third-party social media APIs (Facebook, Instagram, Snapchat) are subject to policy changes, rate limits, and unexpected deprecation. Mitigation: Design the social integration layer as an abstracted service module, so that any individual API can be disabled or replaced without affecting core functionality. Implement graceful degradation — the app remains fully functional even if social accounts are not linked.

Risk 2: Student Data Privacy & FERPA Compliance

Handling student identity information and social media data introduces legal and ethical responsibilities, including potential FERPA considerations. Mitigation: Limit the scope of collected data to only what is necessary for platform features. Clearly communicate a privacy policy to users. Consult UMass IT policy guidelines and use OAuth to avoid storing raw social media credentials.

Risk 3: Scalability Under Peak Demand

Usage will likely spike at the beginning of each semester when students are actively searching for housing. Mitigation: Design the backend with stateless REST APIs to allow horizontal scaling. Use a cloud-hosted database (e.g., PostgreSQL on Heroku or AWS RDS) that can be scaled independently of the application tier. Conduct load testing before each anticipated peak period.

Risk 4: Timeline & Team Coordination

A team of four students balancing multiple courses may face scheduling and workload challenges, particularly around midterms and finals. Mitigation: Adopt a sprint-based workflow with weekly stand-ups, clear GitHub issue assignments, and a shared Google Drive document for async updates. Define a minimum viable product (MVP) scope early to ensure core features are delivered even if stretch goals are deferred.